Peter Yousefi

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SUMMARY

Computer science graduate (June 2026) focused on GenAI and AI engineering. Builds LLM-powered and agentic systems that are measurable, traceable and kept under human control. Brings a background in payroll data accuracy and teaching, with a habit of explaining why something failed.

TECHNICAL SKILLS

AI / GenAI: LLM applications, OpenAI API, Azure OpenAI, structured-output validation, evaluation harnesses, citation validation, agent observability
Languages and Frameworks: Python, TypeScript, JavaScript, SQL, FastAPI, Fastify, Flask, Node.js, React, Next.js, Pydantic, Zod
Data, Cloud and Testing: PostgreSQL, Prisma, SQLAlchemy, Azure (Container Apps, Service Bus, Storage), Docker, GitHub Actions, OpenTelemetry, pytest, Vitest, Playwright

PROJECTS

Agent Flight Recorder | Execution control plane for AI agents — Live demo

- Agent jobs can fail after queuing, spend budget before timing out, or run twice after a crash, and leave no record of why. Built a control plane that stores every attempt, event and cost, so a failed run can be inspected, retried or replayed without losing its history.
- Added retries with backoff, dead-letter handling and per-run budget limits, and traced each request from API to worker. Backed by 195 automated tests.

CrawlOps | Reliability checks for AI web-research agents — Live demo

- A research agent can return a successful response with an empty, malformed or poorly sourced answer. Built an evaluator that grades every run against a written contract using seven deterministic checks, labelling it success, partial or failed with a recorded reason.
- Failures are classified and retries bounded. Reliability trends come from stored runs, so reviewing them costs zero API calls.

AgentDrift | Detection of risky AI-agent data movement — Live demo

- Actions that are each permitted can still add up to data exfiltration. Built detection from statistical baselines and a 15-minute window that links reads, staging and sends, with every report citing real events and a human approving any simulated response.
- After testing on 380 held-out synthetic runs, revised the rules: precision rose from 45.5% to 71.4% and false alarms fell from 60% to 20%, with recall unchanged at 55.6%. Published the results as measured.

AegisOps | Human-approved AI incident response — GitHub

- On-call engineers piece together alerts, logs, deploy history and runbooks under time pressure. Built a workflow that gathers that evidence, proposes a fix with cited sources, and waits for approval before any simulated action.
- Model output is treated as untrusted: strict schemas reject bad responses and citations must match real records. Every step goes to an append-only audit trail.

Continuity | Consistent-character storyboard generator — Live demo

- Image generators drift, so a character looks different in every frame. Built an app that splits a story into up to six scenes, adds a style guide and character sheet, and lets users edit prompts first; a side-by-side mode shows what the character sheet contributes.

Also: **Falsifier** (vets planet candidates through seven tests) | **Lumina** (event-driven campaign generator using IBM Granite)

EXPERIENCE

Teaching Assistant, Trent University — Sep 2025 – Dec 2025

- Replaced plain right-or-wrong marking with comments on every assignment that name the exact concept behind each mistake.
- Ran one-on-one debugging sessions that taught students to isolate bugs themselves, and wrote study materials for the topics they found hardest.

HRIS Specialist (Intern), Concordia University — Apr 2022 – Sep 2022

- Wrote joined queries against the live HR system to produce the employee and payroll reports its built-in tools could not.
- Checked data replication during an ERP modernization and corrected discrepancies before they reached payroll.

Vice President of Social, SCS Concordia — Sep 2021 – Sep 2022

- Rebuilt the event calendar around member feedback and shared ownership across an 8-person committee, lifting active participation by 40%.
- Won “Event of the Year” twice among 12 societies; ran logistics for 200+ attendee events with administration and vendors.

EDUCATION AND CERTIFICATION

Bachelor of Computer Science, Trent University — 2023 – 2026
Microsoft Certified: Azure Fundamentals | View credential